YIFEI SUN

Senior Cloud Software Engineer · Java & AWS Cloud Platform Infrastructure · 13 Years

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SUMMARY

Senior cloud platform engineer with 13 years building large-scale distributed services and cloud-native infrastructure in Java on AWS. Deep expertise in core Java (Collections, Streams, Concurrency, I/O), REST API and gRPC design, and cloud infrastructure automation (Terraform, Ansible, SQS/SNS, ECS/EKS). Led cross-team redesign of a high-traffic API serving millions of daily requests (p99 250ms→50ms, availability 98%→99.9%), deploying on AWS with Kubernetes orchestration. Established cloud architecture standards and observability practices adopted across 10+ teams and 3 departments. Combines deep hands-on Java engineering with cloud infrastructure expertise and cross-organizational technical leadership.

LEADERSHIP HIGHLIGHTS

- Led cross-team cloud API redesign on AWS: p99 5× reduction, 99.9% availability, Kubernetes-orchestrated microservices
- Built cloud infrastructure from scratch — AWS provisioning (Ansible/Terraform), SQS/SNS event pipelines, ECS/EKS deployments
- Drove cloud architecture and observability standards across 10+ teams, 3 departments via RFCs and design docs
- Mentored senior engineers to Principal-level scope; built shared cloud platform services and internal developer tooling

Technologies: Core Java (Collections, Streams, Concurrency, I/O), Spring Boot, Python, Node.js · **AWS** (EC2, ECS, EKS, SQS, SNS, S3, RDS, DynamoDB), Terraform, Ansible · Kubernetes, Docker, Kafka, Redis, PostgreSQL, Cassandra · REST API, gRPC, GraphQL (Netflix DGS) · Microservices, Event-Driven, CQRS · Datadog, Grafana, Distributed Tracing · CI/CD

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead

Oct 2019 – Present

High-Traffic Cloud API Modernization — AWS Microservices

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core API behind CarGurus' most-visited page (~8,000 rpm, millions of daily impressions), deploying on AWS with Kubernetes-orchestrated microservices.
- Introduced a new orchestration service consolidating 12 downstream integrations into 5 well-defined cloud services with circuit breaking, fault isolation, and idempotent retry, reducing fan-out and simplifying failure modes.
- Built automated parity-testing infrastructure comparing new vs. legacy responses at 50% production traffic, enabling safe, data-driven rollout decisions.

  p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)
- Defined cloud architecture standards and observability requirements for microservices; instrumented with structured logging, distributed tracing, and SLO-based latency budgets per downstream dependency.

GraphQL Federation — Delivery Lookup DGS Subgraph

- Designed and integrated a Netflix DGS subgraph into CarGurus' federated GraphQL gateway, exposing vehicle delivery data to frontend, backend, and AI consumers from a single graph.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies to avoid JIT compilation latency spikes on cold starts.

Digital Commerce Platform — Cloud-Native Event-Driven Architecture

- Led backend design for a new commerce product on AWS using microservice and event-driven architecture (SQS/SNS for async messaging, Kafka for event streaming) with Terraform-managed infrastructure.
- Built infrastructure for rapid product experimentation — enabling the company to quickly launch, measure, and safely retire product bets without destabilizing core systems.

Cross-Org Cloud Platform Leadership & Standards

- Established patterns for REST API design, cloud observability, caching, and distributed orchestration adopted across multiple teams; authored design docs, RFCs, and decision records.
- Led 10+ AWS training sessions for engineering teams. Mentored senior engineers on cloud architecture, system design, and path to Principal-level scope.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Modernization

- Upgraded legacy authentication to a modern Keycloak-based SSO platform on AWS, unifying auth across internal systems, an enterprise partner platform, and a large external community. Migrated multiple Java and Node.js services behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Initiative

- Drove early design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Cross-Org Collaboration

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations with clear observability and rollback paths.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Monolith → Microservices Decomposition

- Among the earliest engineers during hypergrowth; designed and implemented core cloud services end-to-end (Order Service, Labor Service, foundational REST APIs) that became the basis for future platform growth.
- Set up the initial AWS environment using Ansible — provisioned infrastructure, deployment pipelines, and monitoring before a dedicated DevOps team existed. Created the first automated testing framework, enabling safer releases during rapid product expansion.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners (e.g., workforce management), ensuring reliable bidirectional sync and schema evolution. Partnered with PMs, sales, and support to translate early customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools (Java backend) for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010